

MDM
Universidade de
Coimbra

2nd Semester
2025/2026

Interface Technologies Project

The Interface Technologies project (course year 2025/2026) has been divided into five main milestones, each one with a deadline. It is mandatory to be developed by groups of **three students** (attending the same PL class).

Students shall create a physical tailor-made user interface or artifact for humans, relying on their knowledge about interaction technologies. The developed product should comprise two parts: (i) *a physical interface component*, using an Arduino; (ii) *an application component*, written in any suitable programming language for your project requirements. The application component should work in cooperation with the physical computing components, exchanging sensor(s) and actuator(s) information with the Arduino (via serial communication, or other means).

Minimum requirements for the project:

- At least 2 *types of sensory physical inputs* (e.g. movement, light, temperature, sound, etc.)
- At least 1 *type of physical output* or actuator
- Integration of, at least, *one capability related to sound or image processing* in the application component

The product should address one theme from the following list.

1. Interface technologies for elderly;
2. Interface technologies for people with special accessibility needs;
3. Interface technologies for children and learning;
4. Co-creative and cooperative computing on design;
5. Interface technologies for new media art;
6. Interface technologies for promoting climate awareness;
7. Interface technologies for urban spaces and physical environments;

Students interested in developing work that does not address any of the proposed themes must seek permission from the professors before the first milestone of the project.

MILESTONES

The project is sub-divided into **five milestones**. Submissions regarding each milestone will be submitted on Inforestudante. Each milestone shall be delivered until **23h59m** of the deadline day. Deliveries after the scheduled day and time will imply a 25% penalty.

Milestones 1, 2 and 3 will be subject to a defense period, scheduled via Inforestudante (each group must register/enroll for its own slot). This period will take place in the PL classes following the submission of the milestone. Students should prepare a visual presentation for the defense.

Milestone 0 — Group and theme definition (22 February) Text file identifying the selected theme and the three group elements (full names and student numbers), to be submitted on Inforestudante.

Milestone 1 — State of the Art and Theoretical discussion (8 March) Written essay presenting the state of the art of the chosen topic and a critical overview on the advantages and disadvantages of the presented approaches. We expected you to present your opinion.

The essay should have 12 to 15 pages (*c.* 6500 words), using a Single-column template, Georgia typeface at 11 pts (leading: 15 pts). Cover and front pages, table of contents, and bibliography are excluded from the page count.

Milestone 2 — Conceptualisation and Prototyping (12 April)
Conceptualisation and early prototyping of the product. This milestone comprises two deliverables:

1. A report (*c.* 5000 words) describing the concept, the steps of the interaction design methodology involved, and the implementation strategy (including a simple description of the plan and analysis/identification of critical issues that might obstruct the development). We expect you to provide as much prototyping detail as possible (*e.g.* 3D sketches, 2D projections, sketches, or high-fidelity prototypes of any object that you intend to build).
2. A partial prototype implementation 25-50% of the final product.
Grading for this milestone will take into account the development stage. The Arduino part of the implementation should include a schematic for the circuit and the corresponding code.

Milestone 3 — Proof of Concept (17 May)
Final prototype (physical implementation, with the full code and schematics for all Arduino and Application components).

Milestone 4 — Documentation (7 June)
Documentation of the developed product. This milestone comprises three deliverables:

1. A report (*c.* 7000 words) for the complete concept. It must include the information from the second milestone report, refined and updated accordingly with the final development of the concept. Schematics and diagrams for the final version of the implementation must be included;
2. A public repository (preferably GitHub) with the documentation and code of the project;
3. Image (min: 3 images) and video (min: 1 video, 720p, 1m30s) documenting the product.

EVALUATION

The evaluation of this project comprises 10 points of the final course evaluation (50%). Students should obtain the minimum classification of 3.5 points out of 10 (35%) in the project evaluation in order to pass the course.

The evaluation of the project is calculated in the following way:

Milestone 1: 3 points out of 10 (30%);

Milestone 2: 2 points out of 10 (20%);

Milestone 3: 4 points out of 10 (40%);

Milestone 4: 1 point out of 10 (10%).

Each milestone will be evaluated based on the following criteria:

Milestone 1: Relation of the essay with the theme; Quality of the research and references; Writing; Presentation;

Milestone 2: Concept complexity; Design methodology; Documentation; Presentation;

Milestone 3: Project Materialisation; Diversity and quality of resources involved; Demonstration; Presentation;

Milestone 4: Documentation.